

Work/Activities

1. Microbiology Undergraduate Researcher

Category:
Research/Lab

In Dr. Alejandra Yep's lab, I investigate iron acquisition and motility in uropathogenic *E. coli* as a potential therapeutic strategy for treating urinary tract infections. Since my first year, I have performed bacterial culture work, growth curves, motility assays, literature review, and data analysis. As my responsibilities increased, I trained new undergraduate researchers in laboratory protocols and troubleshooting. I later began pursuing more independent projects, including investigating surfactant effects on motility and developing software to improve experimental workflows for other research laboratories at Cal Poly.

MME

I remember vividly the many times I had to re-perform growth curves over an entire summer. Wells dried up; controls became contaminated; strains that were supposed to grow simply refused to. Additions to the protocol throughout the summer often meant repeating experiments I thought were finished. I was—generously put—somewhat irritated, to the point that I occasionally blamed my own failures on the innocent bacteria (and yes, I am ashamed of this). Research already seemed elusive, with no clear path between the start and the goal, and its repetition only made the uncertainty more frustrating.

And yet Dr. Yep's patience, and even more so the tenacity I observed in my peers, changed how I understood research. Often, it is tedious, sometimes even menial, but the persistent attempt despite one's own frustration is what propels science forward. This realization brought my corroding relationship with bacteria (we're still working things out) into the research facility's reading room, where I explained my problems to peers from other labs. Their questions interrupted my usual ways of thinking and gave me new approaches to old problems. In turn, I found myself doing the same for them. Research became less solitary, less procedural, and more a shared effort to think carefully through uncertainty.

2. Chemistry Undergraduate Researcher

Category:
Research/Lab

In collaboration with Dr. Corrine Lehr's lab, I worked to quantify intracellular iron concentrations in isogenic *E. coli* strains using atomic absorption spectroscopy. Since our project relied on assumptions about intracellular iron levels, we sought empirical evidence under the conditions we were interested in. I studied the mechanisms and limitations of AAS while helping develop protocols for sample preparation and analysis. This experience strengthened my interest in

analytical chemistry and taught me how science involves constant questioning of one's own assumptions.

3. Posters & Scientific Presentations

Category:

Posters/Presentations

I presented my research at ASM 2025, CSU BIOTECH 2026, WCBSURC 2025, and college-wide undergraduate research conferences in 2024, 2025, and 2026. I appreciated undergraduate conferences for creating an open environment where students from different disciplines could freely ask questions and learn from one another without fear of judgment. Presenting later at ASM exposed me to researchers with deeper expertise who challenged our work with pointed questions. Though humbling, these experiences strengthened both my scientific communication skills and my curiosity about research.

4. Clinical Assistant

Category:

Paid Clinical Experience

As a Clinical Assistant at the Cal Poly Health Center, I supported student patient care by taking vitals, rooming patients, cleaning exam rooms, collecting urine samples, and transporting injured students back to their dorms or apartments. This role taught me how much of good care depends on small, meticulous, but meaningful actions. I remember assisting a student who was crying in the hallway while holding an emesis bag. Instead of rushing the rooming process, I told her to take her time and helped her slowly to the exam room. Through this experience, I learned that patient comfort often begins even before the physician enters.

5. Hospice Volunteer

Category:

Community Service/Volunteer – Medical/Clinical

As a volunteer with Hospice SLO County, I have visited an elderly hospice patient biweekly since the summer after my first year to provide companionship and support during end-of-life care. Even after her transition into a senior home, I continue visiting her regularly. Working with a patient experiencing dementia taught me that communication often depends less on direct conversation and more on careful observation. Over time, I became more aware of how consistency and patience can impact a patient's quality of life.

MME

Frankly, Beverley was not the client I expected to volunteer for. During training, I had requested to work with seniors capable of coherent conversation. So, as one might imagine, I was much surprised to meet Beverley for the first time.

She could respond to simple questions, but anything more seemed beyond her cognitive capabilities. Her gaze was perpetually distant, as if searching for something absent. Hoping to build a strong relationship with hospice clientele, I was—though I now find this ridiculous—somewhat disappointed. I tried speaking with her as much as I could, to get to know her as much as I could. But this often tired her out, as did the physical therapy exercises we did together despite her reluctance. In retrospect, I realized I might have imposed myself upon her.

After many visits, I started noticing small habits in her language and movements: the way she limped, almost unnoticeably, on one leg; the way she asked, elusively, to go on walks; the way she stared out onto the street. Through these seemingly meaningless observations, I found I could get to know her beyond what could be said with words. Hospice taught me that care does not always begin with conversation or even understanding; sometimes it begins by learning how another person communicates before deciding what they need.

6. Restorative Partners Volunteer

Category:

Community Service/Volunteer – Non-Medical

After completing restorative justice training with Restorative Partners, I began visiting juvenile hall weekly to support the incarcerated youth through playing sports, cards, and casual conversation. These visits became a consistent way to develop genuine camaraderie and learn about their experiences, many of which involved poverty and limited opportunity. Over time, I became more aware of how strongly social conditions shape behavior. This experience has made me more conscious of my own privilege and strengthened my interest in working with underserved populations in medicine.

MME

There's often an unnamed tension as new volunteers stream in. It is as though the incarcerated youth tacitly signal: I did not ask to be helped. And this is natural; what could privileged individuals from outside understand about their experience? The first time I volunteered, these conflicts rattled my mind. If I was perpetuating some paternalistic agenda, there was no point in me being there.

Yes, they have committed crimes—some ghastly. Yet fundamentally, they are my peers in so many surprising ways. I learned that speaking casually with them—often while sharing food or playing cards—develops trust between people whose experiences are nothing alike. Expressing the willingness to learn about their lives, thoughts, and feelings led to friendships worth remembering. Many come from worlds I have never had to inhabit, cursed by poverty and failed

institutions. Getting to know them led me to discover their love of art, especially lyrics and poetry, and their willingness to share it. After several discussions with staff, I convinced the facility to publish their art in weekly newsletters sent to donors, volunteers, and guardians.

I want to work with underserved populations in medicine because care begins not with the presumptuous assertion that I understand, but with remaining present long enough to try.

7. Learning Assistant

Category:

Teaching/Tutoring/TA

As a Chemistry Learning Assistant, I worked with faculty to facilitate problem-solving in a general chemistry classroom. My role was not to simply give students answers, but to help them reason through where their understanding had failed, one-on-one. I supported students during class, graded assignments after class, and prepared weekly for prospective lab experiments with other learning assistants and instructors. Teaching made me receptive to how different students learn in different ways, and how patience, trial-and-error, and empathy are essential for building a welcoming learning environment.

8. Physician Mentorship / Shadowing

Category:

Physician Shadowing/Clinical Observation

- involved shadowing in nephrology, pediatrics, neonatology, orthopedic surgery, and anesthesiology. rotated through the different specialties for an entire summer. observed interactions with patients; learned that asking open-ended questions is good to gauge patient history and build relationships. actively brought questions to the clinic to discuss with the physician to further own knowledge. witnessed procedures like knee replacement, hip replacement, and a full delivery. I was astonished at how each person in the OR had their own little role and how they worked like one big machine; very collaborative.

Through the Dignity Health Physician Mentorship Program, I rotated through nephrology, pediatrics, neonatology, orthopedic surgery, and anesthesiology over a summer. I observed clinic visits, patient histories, a delivery, knee and hip replacements, and perioperative care. I brought questions to clinic to discuss with physicians to better understand their reasoning. It was intriguing to see how open-ended questions help physicians learn a patient's story while building trust, and how operating rooms function through many small, coordinated roles rather than a single person's work. The experience overall has consolidated my interest in medicine.

9. ASI University Student Governor

Category:

Leadership

As an ASI University Student Governor on the University Union Advisory Board, I represented students in decisions affecting ASI-managed spaces, including the University Union and Recreation Center. During my term, I identified the lack of menstrual products in student-government-run facilities and worked with two peers to research, propose, budget, and implement sanitary pads in campus restrooms. This project—which at first seemed distant and unrealistic—made student advocacy feel much more attainable. A basic need still required planning, budgeting, and coordination before it could become an actual change in the spaces students use every day. Overall, it was quite a bit of fun working with my fellow ASI members to achieve a common goal.

10. BCSM Ambassador

Category:

Leadership/Community Outreach

As a Bailey College of Science and Mathematics Ambassador, through all three years, I gave quarterly tours, participated in outreach volunteering, and represented the college at events such as Open House, where I spoke on a panel to 900 prospective students and guardians. My role was to explain Cal Poly's academic, research, and extracurricular opportunities through both institutional knowledge and my own experience as a student. This work allowed me to make the large university feel more welcoming for students wavering on where they might belong. Through this experience, I learned that outreach depends on making the path ahead feel understandable by breaking it down into smaller chunks.

11. University DEI Committee

Category:

Leadership/Advocacy

As a student member of the Bailey College Committee for Inclusion and Equity, I worked with faculty and students to identify barriers to student success and support a more inclusive college culture. I participated in prospective-student panels, helped plan DEI-related events, and contributed to conversations about how policies and practices affect access and belonging. This experience taught me that equity work often involves open-ended survey work, speaking directly with students while listening carefully, recognizing where systems fail students, and working through institutions to eliminate those failures.

12. Frost Fellow

Category:

Honors/Awards/Scholarships

As a Frost Research Scholar, I receive \$10,000 annually to continue research through the remainder of my undergraduate career, along with a \$4,500 summer research stipend. The scholarship changed the conditions of my research. That is, it gave me time to stay with questions that refused to resolve quickly and to work toward projects with enough depth for publications. It is not simply an award, but more a responsibility to use the support toward continued scientific work.

13. Scientific Software Development

Category:

Other / Research

After repeated conversations in lab about how manual swim-diameter measurements and colony counting slowed data analysis, I developed AgarLens, a desktop app for analyzing agar plate images. The program measures growth diameter and area from scanned plates, counts bacterial colonies using a YOLO model, and exports annotated images for analysis. Making the tool allowed me to treat a laboratory inconvenience as a fun computational problem. It also gave the project a use beyond my own data, since the code is public and can be used or modified by other labs. The project can be found here: <https://github.com/jinsko1/AgarLens>, and I am planning on officially publishing it through a software journal.

14. Academic Website / Educational Resource Development

Category:

Other

I created and maintain a personal academic website (jinsko.com) as a way of genuinely understanding what I am doing in the lab rather than simply following procedures. Writing about the projects forces me to explain the background, mechanisms, and questions the assumptions behind my work, which often exposed gaps in my understanding, leading to more literature review. The site also became a place to study topics I was independently curious about: bacterial antagonism, sham surgeries, and software development. That is, the process made learning more deliberate. If I could not explain an idea clearly, I did not understand it well enough.

15. Hobbies / Personal Interests (Optional but Recommended)

Category:

Hobbies/Avocations

As a hobby, I read widely, from inexpensive mass-market paperbacks to textbooks, though I am most drawn to novels. Beyond the pleasure of reading itself, fiction offers me a practical benefit: it lets me inhabit minds unlike my own. In that exchange of consciousness, the boundaries of the self become less rigid, more porous. Reading has made me more comfortable with ambiguity, more attentive to perspective, and more receptive to different ways of thinking. Among some of my favorites are *Great Expectations*, *Frankenstein*, *1984*, and *Bel Canto*. For medicine specifically, I would recommend works by Atul Gawande. Any article/textbook by David Sackett is fantastic.
